

# Aditya Pandey

adityaadityapandey2002@gmail.com | +91 8178525793 | [LinkedIn](#)

## Education

**Bachelor of Technology:** Artificial Intelligence & Machine Learning USICT – GGSIPU, Delhi— 2025  
**Senior Secondary (CBSE):** PCM + Computer Science (C++) – 95% Bal Bhavan Public School, Delhi — 2020

## Technical Skills

**Languages:** C++, C#, .NET, Java, Python, SQL.  
**Technologies:** MERN, AI/ML, Data Analysis, Distributed Systems, Docker, Git, CI/CD, Linux, Redis, AWS.  
**Relevant Courses:** Operating Systems, DBMS, OOP, DSA, System Design.

## Achievements

- Secured Top **1%** Rank (among 100K+) in CodeKaze Coding Contest 2024. [Contest Rank Certificate \(Top 1%\)](#)
- Achieved a Countrywide Top **5%** JEE rank among 1M+ applicants and **1st** in School in XII Exams 95%.
- Ranked among Top **6%** in GATE 2024 out of 1L+ aspirants nationwide.
- Codeforces Specialist with 900+ DSA problems solved on LeetCode and multiple coding platforms.
- Oracle Certified Professional – Multicloud Architect (OCI-AWS-GCP) and AI Workflow Automation [Certificate](#)

## Experience

**Tata Consultancy Services** Sep 2025 – Nov 2025  
Backend System Trainee

- Accelerated backend performance by 2× via SQL tuning, caching, and resilient system design using Java and Spring Boot.
- Created high-throughput microservices for scalable, fault-tolerant systems using Docker and CI/CD.
- Engaged in hackathons and engineering knowledge sessions, elevating cross-team collaboration and innovation across projects.

**Opkey (Product-Based SaaS Company)** Jul 2025 – Sep 2025  
AI/ML Python Backend Developer – R&D

- Engineered LLM-driven automation with RAG and enhanced system prompts, increasing agent reliability by 45%.
- Used structured logging for LangChain workflows, reducing tool-call errors by 35% and enabling 4× faster debugging.
- Optimized Opkey’s deployed systems via Nginx, lowering downtime by **20%** and improving routing throughput by **30%**.
- Automated SAP/Oracle test workflows, cutting manual intervention by 70% and improving test-cycle accuracy by **40%**.

**Prodigal AI** Feb 2025 – Jun 2025  
AI Software Developer

- Led a team to integrate multi-platform agent APIs and improved TCP/IP streams, increasing throughput by 40%.
- Improved search relevance and IR efficiency by 60% via LLM-powered metadata pipelines for distributed indexing systems.
- Published a production-ready, **open-source** CLI Twitter automation tool on PyPI, supported by Git-driven workflows.

**Kaizenspark** Dec 2024 – Feb 2025  
Lead Full Stack Developer

- Directed a team of 5 developers to build scalable mobile-friendly marketplace app with React, Node.js and CI/CD.
- Mentored peers and advanced code quality that cut down costs by 25% while reducing defects by 40%.

**IBM** Jun 2024 – Aug 2024  
Data Analyst Intern

- Delivered anomaly detection systems, improving forecast accuracy by 30% and ingestion speed by 25%.
- Analyzed food distribution data via Python and SQL, driving impact for **8K+** underserved residents.
- Applied lightweight data compression to speed up processing and cut storage by 20%.

## Research Work

**Dynamic Analysis of CCTV Feed via ML Algorithms** Jul 2023 – Nov 2023  
Guide: Prof. Ashish Payal, USICT Delhi [Research Thesis](#) | [Demo](#)

- Implemented adaptive services leveraging blink-to-eye and smile-width analytics for predictive state recognition.
- Developed a CNN-based facial landmark detector with GPU-level parallelism (30% faster, 18% lower memory).
- Applied statistical modeling to convert visual cues into time-series datasets for anomaly detection with **92%** accuracy.
- Technologies: Python, OpenCV, Dlib, CNN, ML, PuLP, Facial Landmark Detection, Statistics & Mathematical Modeling.

## Projects

**AI Personal Assistant “Rockerz” (NLP + CV + Speech Recognition)** [Demo](#) — Jan 2025 – Jun 2025  
Guide: Prof. Vandana Nath, USICT Delhi

- Built “Assist AI”, a Python assistant with CV, NLP & Brain-Computer Interface for 75% accurate hands-free automation.
- Designed analog signal filters that improved EEG signals accuracy by 65% and reduced system noise by 50%.

**Real-Time Object Classifier – OCR Engine (Machine Learning)** [Demo](#) — Oct 2024 – Dec 2024

- Trained a custom YOLOv8 model on 1.2K+ labeled images (fruits/vegetables) attaining F1: **0.92** and **97.5%** recall.
- Boosted real-time processing speed by 20% through targeted model fine-tuning for handling items at large-scale industries.
- Achieved 95% OCR accuracy on 500+ samples via adaptive filtering and iterative model refinement.

**Web Gallery Application (Image Gallery)** [Live](#) — Aug 2024 – Sep 2024

- MERN-based image gallery using Model-View architecture supporting **1K+** uploads via CRUD for interactive collages.
- Streamlined UI using React, TypeScript, Tailwind CSS, MongoDB, Postman with lazy loading to improve load time by **50%**.